

The Procurement Problem

On the Structural Properties of Questions That Can Reach What We Don't Know We Don't Know

Nicholas Jacob Bogaert

AI.Web Research • March 2026

nicbogaert@aiwebhq.com • aiwebhq.com

I. The Problem That Isn't Named

There is a failure mode so common it has become invisible. We call it not-knowing, but that description is too generous. What we actually face, most of the time, is something more specific and more treacherous: *not knowing that we don't know*.

The unknown unknown. Donald Rumsfeld's infamous press conference gave it a name in 2002, and the phrase caught because it pointed at something real. But naming the category didn't help us reach it. We know now that unknown unknowns exist. We have no systematic method for procuring them. The gap between knowing the territory exists and being able to navigate it has remained, more or less, intact.

This paper is about that gap — and more specifically, about one part of it that has received almost no attention: the structure of the question itself.

We have sophisticated epistemologies. We have research methodologies, structured inquiry frameworks, intelligence tradecraft, scientific method, Bayesian updating. All of these assume you already know enough to ask the right question. They are instruments for drilling down from a known question to its answer, for refining hypotheses you've already formed, for testing predictions you've already made. They are superb at procuring known unknowns — the things you know you don't know, where you can articulate the gap clearly enough to direct a search toward it.

None of them, structurally, can reach an unknown unknown. Not because they're badly designed. Because *their design is the obstacle*. The more precisely a method specifies what counts as a valid answer, the more completely it forecloses the possibility of encountering something that falls outside those specifications.

This isn't a solvable problem at the level of technique. It's a structural feature of how questions work. And if we want to do anything about it, we have to start at the beginning — with what a question actually is, what it does to the space of possible answers before any answer arrives, and what it would mean for a question to leave that space genuinely open.

II. What a Question Actually Does

Start with something so obvious it's easy to miss: a question is not neutral.

When we ask a question, we don't simply point at an empty space and wait for it to fill. We do something more specific. We constitute a frame. The question establishes which kinds of answers are possible, what would count as relevant evidence, which prior knowledge applies, and — critically — what would be recognized as an answer at all if it arrived.

R.G. Collingwood, the British philosopher and archaeologist, made this the cornerstone of his epistemology. From his excavation work, he arrived at a principle that sounds obvious once stated but whose implications are radical: knowledge comes only by answering questions, and those questions must be the right questions asked in the right order. More than that: the truth or falsity of a statement is not a property of the statement in isolation. It is a property of the relationship between the statement and the question to which it is an answer. A proposition that is true in response to one question can be meaningless — not false, but *meaningless* — in response to another, because it isn't answering the question that was actually asked.

This means the question does not merely point toward an answer. The question *constitutes* what an answer is. It defines the shape of the space before anything occupies that space.

Hans-Georg Gadamer, working in the German hermeneutic tradition, reached the same territory from a different direction. In *Truth and Method*, his account of the "genuine question" makes a distinction that most epistemology ignores. A genuine question is one that leaves its answer genuinely open — not bounded by what the questioner already expects to find. "The openness of a question is not boundless," Gadamer writes. "It is limited by the horizon of the question." But a question that lacks this horizon at all — a question asked with a predetermined answer in mind, or one framed so narrowly that only certain kinds of answers can reach it — is, in his terms, not really a question at all. It is a performance of questioning. It produces the form of inquiry without its function.

And its function, for Gadamer, is specific: the genuine question "breaks open the being of the object." It creates the conditions under which something other than what we already know can arrive.

This is not mysticism. It is a structural claim about the epistemological relationship between question and answer, and it has a precise implication: most of what we call asking questions is actually a kind of answering in disguise. We formulate questions in ways that encode the answer we already half-expect, or that filter what counts as a response through the categories we already possess. The question "which of these three options is best?" can only return "option A," "option B," or "option C." The question "is my hypothesis correct?" can only return "yes" or "no." The question "what are the best practices for X?" can only return things that fit inside the concept of "best practice." These questions are not open. They are closed systems wearing the syntax of inquiry.

Against this, consider what an actually open question must do. It must resist premature closure. It must hold multiple possible directions simultaneously. It must, in some sense, not know what it's looking for — because the moment it knows what it's looking for, it has already foreclosed finding anything else.

III. The Indeterminacy Requirement

John Dewey, working in the American pragmatist tradition, gives us the third piece of this structure. Dewey's account of inquiry begins not with a question but with a *situation* — specifically, an indeterminate situation. Not a puzzle you've formulated. Not a gap you've identified. A situation that is inherently unsettled, troubled, ambiguous. Something is wrong and you don't yet know what it is or how to describe it. This felt disturbance — Dewey calls it a "pervasive quality" that colors the entire situation — is the precondition for genuine inquiry.

"Problems do not preexist inquiry," Dewey writes. They are not waiting out there to be discovered and addressed. They are constituted *in the act of inquiry itself*, as we try to articulate what exactly is wrong with the indeterminate situation we find ourselves in. The formulation of the problem is already part of the investigation, not a prerequisite for it.

This matters enormously for the question of unknown unknowns, because it identifies the pathology precisely. Most institutional inquiry begins not with genuine indeterminacy but with a *manufactured* question — a question formulated before the investigation begins, derived from an agenda, a hypothesis, a grant proposal, a research plan. The indeterminacy has been pre-resolved at the level of question formation. You've already decided what the problem is. You're now searching for its solution.

Dewey calls manufactured doubt "pathological." Real doubt, for him, arises from a real situation — something in the environment that actually disrupts the ordinary flow of engagement with the world. Personal states of doubt "that are not evoked by and are not relative to some existential situation" are not the engine of inquiry. They are noise, or performance.

We are now in a position to see why unknown unknowns are structurally inaccessible to most inquiry. The unknown unknown, by definition, is something you haven't yet identified as a problem. It hasn't disrupted your ordinary engagement with the world enough to generate genuine indeterminacy, because you don't yet have the categories to notice it as a disruption. The standard response — manufacture a question and go looking for it — is precisely what Dewey identifies as pathological. You can't find what you don't know to look for by constructing a search for it. The construction of the search embeds the categories that define what you'll find.

IV. The Hierarchy of Questions

Gregory Bateson, the anthropologist and cybernetician, gives us the clearest structural account of why this problem persists. In his theory of levels of learning, Bateson distinguishes between different logical types of learning — and by extension, different logical types of inquiry.

Learning I is about getting answers right within a given context. You learn which lever to press, which response to give, which rule applies. The context itself is not questioned. Learning II is about learning the rules of the context — the meta-pattern that governs which Learning I responses are appropriate. Learning III is something rarer and more disorienting: it is learning about the context that constitutes the Learning II context. It is questioning the frame that defines what counts as an appropriate question in the first place.

Most inquiry operates at Learning I. Scientific hypothesis testing, systematic literature review, structured analytical techniques — these are all Learning I operations. They operate within a given framework of what counts as a valid question, what counts as evidence, what counts as a finding. They do not question the framework itself.

Some inquiry reaches Learning II — methodological reflection, paradigm critique, the kinds of debates that happen at the foundations of disciplines. These are rarer and often contentious, because they threaten the working practices of the entire field.

Learning III is, as Bateson notes, "probably quite difficult and rare, even among humans." It requires questioning the habits of thinking that constitute you as the kind of thinker you are. It is not just asking a different question. It is asking from a different position — one that can see the context that usually generates questions without being fully captured by it.

The unknown unknown lives at Learning III. To reach it, the question must somehow operate at the level of the frame rather than within the frame. It must be a question that puts the questioner's own categorical apparatus at risk, not just a question that searches within that apparatus for missing information.

This is where the Zen koan tradition becomes more than a metaphor. The koan is not primarily a riddle. It is a question-structure specifically engineered to operate at Bateson's Learning III. A koan like "what was your face before your parents were born?" is not asking for a factual answer. It is asking a question that the existing categorical apparatus cannot answer — not because the answer is missing, but because the apparatus itself is inadequate to the question. The purpose is the failure. The breakdown of the apparatus under pressure from the question is the point. What emerges from that breakdown cannot be retrieved, because it wasn't in the retrieval system. It has to be generated under conditions of genuine aporia.

The koan is technology for producing genuine not-knowing in a practitioner who has become too fluent, too fast, too capable of answering questions within the existing frame.

V. The Mechanism: Semantic Multiplicity

We now have the philosophical architecture. What we've been missing — what Gadamer, Collingwood, Peirce, Dewey, Bateson, and the koan tradition all point toward but none fully specify — is the *mechanism*. What, structurally, does a question have to do to hold itself open, maintain genuine indeterminacy, and operate at the level of the frame rather than within it?

Here is our proposal: the structural property that enables a question to reach unknown unknowns is **semantic multiplicity** — the deliberate deployment of words and phrases that carry multiple, potentially incommensurable meanings simultaneously, without resolving that multiplicity in advance.

A question that uses words with singular, well-defined meanings can only return answers within the semantic territory those words map. Ask about "revenue optimization" and you can only receive answers about revenue. Ask about "machine learning models" and you can only receive answers about machine learning models. The question's vocabulary is its horizon, and a vocabulary with no ambiguity is a horizon with no space behind it.

But a question whose words inhabit multiple semantic regions simultaneously does something different. When a word holds two or more distinct meanings — not as noise to be resolved but as a deliberately sustained multiplicity — the question activates multiple regions of the knowledge-space at once. The answer does not come from any single region. It emerges from the interaction between regions, from the interference pattern created when multiple activation fields overlap.

This is not the same as vagueness. A vague question is one that could mean many things and doesn't commit to any of them — it produces noise, not novelty. Semantic multiplicity is precision about imprecision. It is the deliberate holding of two or more specific meanings in tension, so that the answer must navigate between them rather than resolve one and discard the others.

Consider what happens when Charles Sanders Peirce describes abduction — the only logical operation that introduces genuinely new ideas. The "surprising fact" that triggers abduction is not just an anomaly within the existing framework. It is a fact that doesn't fit — that sits at the intersection of multiple explanatory frameworks simultaneously and cannot be cleanly assigned to any of them. The surprise is structural. It comes from the activation of incommensurable contexts. The hypothesis that abduction generates is not retrieved from any single context but synthesized across the tension between them.

This is structurally analogous to what semantic multiplicity produces in a question. The question that holds multiple meanings simultaneously creates, in the space it opens, something like Peirce's surprising fact — not a gap to be filled but a junction at which multiple incompatible frameworks press against each other. What emerges from that junction was not available in any single framework. It could only appear at the boundary.

VI. The Three Operations

Drawing on the philosophical architecture above, we can identify three operations that together constitute the semantic multiplicity technique.

Operation 1: Deliberate Incompleteness at Load-Bearing Points

The question that reaches unknown unknowns systematically refuses to complete itself at precisely the moments where completion would determine the answer in advance. This is not grammatical incompleteness — not broken sentences — but semantic incompleteness at the conceptual joints.

"What are all the secret doors labeled?" is not asking which secrets exist. It is refusing to specify what kind of secret, what kind of door, what kind of labeling. Each of those specifications would collapse the question into a single semantic region. The incompleteness at each joint holds the question across all of them simultaneously.

Simone Weil describes the posture this requires as "waiting, not seeking" — thought "empty, waiting, not seeking anything, but ready to receive in its naked truth the object that is to penetrate it." The deliberate incompleteness is not laziness. It is the structural equivalent of Weil's emptied attention: the refusal to fill the question with the questioner's expectations before the answer can arrive.

John Keats called the same quality "negative capability" — the capacity to remain "in uncertainties, mysteries, doubts, without any irritable reaching after fact and reason." The irritable reaching after fact and reason is precisely what premature completion performs. It resolves the question too early, before the question has had the chance to do its work.

Operation 2: Frame Shifting Rather Than Clarifying

Standard inquiry, when it doesn't get the answer it wants, adds precision. More context, more specificity, more constraint. This is appropriate when the problem is that the question was unclear. It is counterproductive when the problem is that the question was too clear — that it over-constrained the space of possible answers.

Frame shifting does the opposite. Instead of adding words to narrow the question, it replaces the conceptual frame entirely. "Secrets" becomes "secret doors." "Tell me" becomes "let me in on." The content of the question may remain similar, but the frame — the implied relationship between questioner and answer-space, the metaphor through which the question is apprehended — shifts entirely.

Each frame shift re-activates the question in a new semantic region. The accumulated effect of multiple frame shifts is not to narrow the question but to build up a multi-dimensional activation space around it. The answer that eventually emerges has been shaped by all the frames, not just the most recent one.

Bateson's Learning III operates through something structurally similar — the questioning of the context that usually generates Learning II responses. Frame shifting is a question-level implementation of the same move. It doesn't add to the frame. It steps outside the frame and redraws it.

Operation 3: The Meta-Question as Instrument

The most consistent and perhaps most powerful operation is the deliberate use of the meta-question — the question about the questioning itself. "What should I have asked that I didn't?" "What are the doors I don't see?" "What does this framework miss?"

The meta-question does something uniquely powerful: it treats the questioner's own categorical apparatus as the object of inquiry. It takes up residence at Bateson's Learning III, not as an accidental by-product but as the explicit goal. By asking what hasn't been asked, it points at the negative space around the existing frame — the territory that the frame's architecture makes invisible.

This is precisely what Collingwood identifies as the highest philosophical activity: not answering questions, but rendering explicit the presuppositions that make certain questions possible and make others impossible. The meta-question is presuppositional analysis directed at one's own inquiry.

VII. Why the Field Got This Backwards

Everything we have described runs against the dominant direction of the field studying human-AI interaction and question formulation. That field — prompt engineering, query optimization, structured analytical techniques — has developed remarkable sophistication at one task: making questions more precise.

The standard advice: be clear and specific, provide context and constraints, define your terms, specify your format. All of this is excellent advice for retrieving known information. It is the epistemic equivalent of a search engine with good keywords — the more precisely you specify what you're looking for, the more accurately you'll find it.

But precision is the enemy of discovery precisely because it resolves multiplicity before the multiplicity can do its generative work. The more precisely you specify what you're looking for, the more completely you foreclose finding what you didn't know to look for.

The entire field has been optimizing for the wrong variable. Not because precision is wrong — it is exactly right for retrieving known unknowns. But because retrieval and discovery are not the same operation. They require different question structures. Optimizing one degrades the other.

What we are describing is not the inverse of precision. It is a different epistemological operation altogether: not better retrieval, but the creation of conditions under which genuine novelty can arrive.

VIII. What This Makes Possible

The implications extend well beyond human-AI interaction, because the mechanism is older than AI and operates wherever questions are asked.

The researcher who hits a genuine breakthrough is almost never the one who followed the research plan most faithfully. They are the one who noticed something anomalous, let it stay anomalous rather than explaining it away, and held the question open long enough for the anomaly to generate a new frame. This is abduction in Peirce's sense, but it is also semantic multiplicity — the anomaly stayed alive because the researcher didn't prematurely assign it to an existing category.

The therapist who helps a patient reach a genuine insight is not the one who already knows the diagnosis and is working toward confirmation. They are the one who can sit with Weil's empty attention, who can hold the patient's situation as genuinely open, who can ask questions whose answers they don't already know.

The conversation that produces something neither participant expected is not the one governed by agendas and talking points. It is the one where both participants let the subject matter lead — where, in Gadamer's formulation, they allow "the genuine question to break open the being of the object."

In each case, the structural property is the same: the question maintains genuine indeterminacy. It activates multiple semantic regions simultaneously. It does not foreclose what could arrive by specifying too early what should arrive.

IX. The Structure of a Question That Can Reach Unknown Unknowns

We can now give a provisional structural account of what a question must be to reach what we don't know we don't know. Not a method, because a method would itself be a form of premature closure. A description of properties — necessary conditions, not sufficient ones.

It must leave its horizon genuinely open. Not boundlessly open — that is vagueness, not inquiry — but open with respect to its own potential answers. The question must be capable of being surprised by what it finds. This requires that the question not encode its answer in advance (Gadamer, Collingwood).

It must arise from genuine rather than manufactured indeterminacy. The question must be motivated by something actually unsettled, actually disrupted — not by a research agenda that has already specified what the problem is (Dewey).

It must operate on the frame, not just within it. Questions that operate within the existing categorical apparatus can only return what exists within that apparatus. The question that reaches unknown unknowns must, in some way, put the apparatus itself at risk — must question the context, not just the contents of the context (Bateson, koan tradition).

It must hold semantic multiplicity rather than resolve it. Multiple meanings held simultaneously, without premature collapse into a single reading, create the interference condition from which genuinely novel synthesis can emerge. This is the mechanism; the others are the conditions under which the mechanism can operate.

It must be asked from a posture of waiting rather than seeking. This is the hardest property to specify structurally, because it is as much a property of the questioner as of the question itself. But the posture matters because seeking encodes what will count as finding. Waiting leaves that determination to the encounter itself (Weil, Keats).

X. What Remains

This paper has argued for a structural account of a class of questions that currently has no name. It has pointed toward historical and philosophical anchor points from which the territory has been glimpsed, without being fully mapped. And it has identified a mechanism — semantic multiplicity — that provides the first explicit account of how such questions achieve what they achieve.

What remains is considerable. We need a more formal characterization of how semantic multiplicity works in practice — what makes a specific word or phrase generative rather than merely ambiguous, how frame shifts combine across a conversation, how the meta-question relates structurally to the content-level question it reflects.

We need to understand the receiver problem: the question that holds semantic multiplicity open requires a receiver capable of tolerating that openness without prematurely resolving it. This is not guaranteed. Both human receivers and artificial systems have strong tendencies toward disambiguation — toward collapsing multiplicity into the most probable single meaning. The question that can reach unknown unknowns may require, for its interlocutor, something like what Keats called negative capability: the trained capacity to remain in uncertainty without reaching prematurely for resolution.

And we need to understand the ethics. A question structure that bypasses the ordinary apparatus of knowledge retrieval and generates novelty from the interaction between semantic regions is powerful. Power requires responsibility. Not every unknown unknown should be known. Some things are inaccessible for good reasons.

But before any of that, there is the naming.

The procurement problem — how to acquire what we don't know we don't know — has remained unsolved not because we lack the intelligence to solve it but because we have been looking in the wrong place. We have been trying to improve our search strategies when the issue is not the strategy but the question. The question that can reach unknown unknowns does not have a better pointing-at. It has a different structure altogether.

That structure is what we have tried, in these pages, to begin to describe.